Wildfire threat prediction and survival analysis based on WiDSWorldwideGlobalDatathon2026

Group menmber:

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1. The background, motivation and objectives of the project

Project selection Kaggle competition WiDSWorldwideGlobalDatathon2026. The competition task requires contestants to use the observation information of the initial stage of wildfires to predict the probability of an evacuation area being threatened by wildfires in the next 12, 24, 48 and 72 hours. Unlike the traditional binary classification, the project not only judges whether an area will be threatened, but also pays attention to the time and risk level of the threat, which is more suitable for analysis from the perspective of events.

The wildfire warning scene is very sensitive to time. For the government, firefighting and evacuation management departments, it is not enough to know whether a region will be threatened in the end. The key is to judge which areas are at higher risk, which areas will be affected earlier, and the probability of threats at different times. Compared with traditional binary classification, survival analysis is more suitable for this task, because it can deal with the time of occurrence of events, risk sorting and deletion of data at the same time, which is more in line with the needs of real emergency decision-making.

The goal of the project is to complete data cleaning, exploratory analysis, visualisation and predictive modelling based on the competition data, build a model that can output the probability of 12h, 24h, 48h and 72h wildfire threat, and compare the performance of different survival analysis methods. According to the results submitted by Kaggle, the prediction effect, advantages and limitations of the model are analysed, and the future improvement direction is further put forward.

2. Description of the competition and data set

According to the description of the competition page, the competition needs to predict the probability of threats in the evacuation area within 12, 24, 48 and 72 hours based on the observation data of the initial stage of the wildfire. The score of the competition takes into account both the risk sorting ability and the probability calibration ability: the overall score is composed of C-index and WeightedBrierScore, and the 48-hour Brier has the highest weight. We understand the task as time to event prediction, not the traditional single-label classification.

Based on the statistical results of train.csv, test.csv and train_augmented_D.csv, the data overview is shown in the following table.

Indicator	Value
Training set sample size	221
Test set sample size	95
Number of original modeling features	34
Number of enhanced modeling features	38
Samples with threat occurring within 72 hours	69(31.2%)
Samples without threat occurring within 72 hours (right-censored)	152(68.8%)
Median time_to_hit_hours	43.11 hours

Maximum observation duration	66.99 hours

Table 1 Overview of Data Set (based on train.csv, test.csv and enhanced feature file statistics)

Judging from the distribution of labels, a total of 69 samples in the training set were threatened within 72 hours, and 152 samples were not threatened during the observation period, and the right deletion rate reached 68.8%. It shows that the data is unbalanced, and there are obviously more samples without events. Statistics found that out of 69 samples of threats, 49 appeared within 12 hours, 63 appeared within 24 hours, and 66 occurred within 48 hours. The results show that early risk identification is crucial.

Main File	Purpose
00_metric_validation.ipynb	Replicates the local scoring function and validates the calculation logic of hybrid metrics.
01_data_clean.ipynb	Performs data inspection, deduplication, handling of missing/infinite values, and exports cleaned data.
02a_baseline_Cox_submission.ipynb	Builds the CoxPH baseline model and generates the first version of the submission file.
03_feature_EDA_B.ipynb/03b_EDA_Visualizations_B.ipynb	Conducts exploratory data analysis (EDA), organizes charts, and analyzes relationships between key features.
03_feature_engineering_D.ipynb	Constructs 4 business-driven new features.
04_model_training.ipynb	Compares the in-sample performance of 9 types of survival models and selects candidate models.
05_hyperparameter_tuning_and_ensemble.ipynb	Performs hyperparameter tuning using Optuna, and exports three types of ensemble submission

	files based on mean/median/rank strategies.

Table 2 The corresponding role of the main workflow in the report

3. Data cleaning, exploratory analysis and visual discovery

In the data processing stage, the structural inspection of train.csv and test.csv was completed first. In Notebook01, uniformly replace the inf/-inf that may exist in the numerical column, check the duplicate samples, and use the training set statistics to fill in the potential missing values. The exported train_clean.csv and test_clean.csv have no residual missing values, which is convenient for subsequent models to read directly. For Cox, SVM and other models that are sensitive to feature scales, StandardScaler is used to standardise in the modelling stage to ensure the stability of parameter solution.

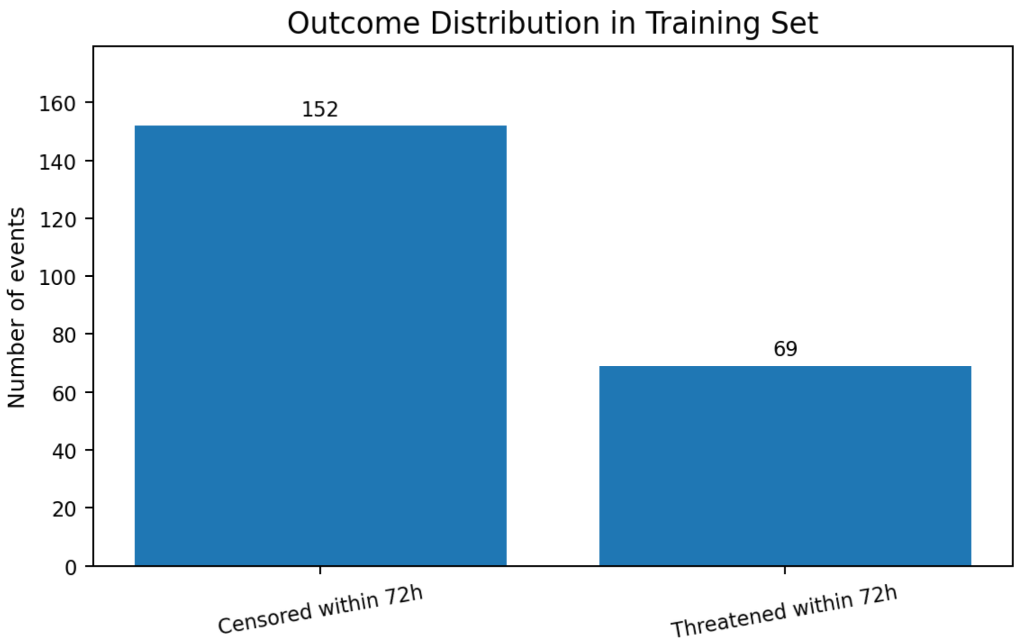

Figure 1 Distribution of training set results: the right deletion sample is obviously more than the threat sample

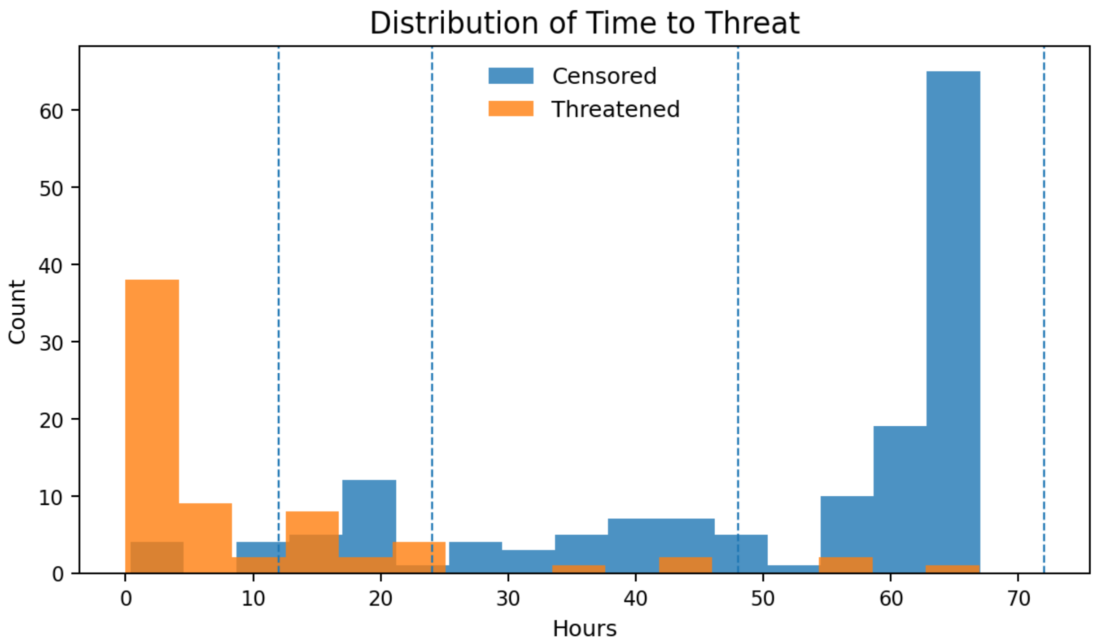

Figure 2 time_to_hit_hours distribution: Most threat samples appear within the first 24 hours

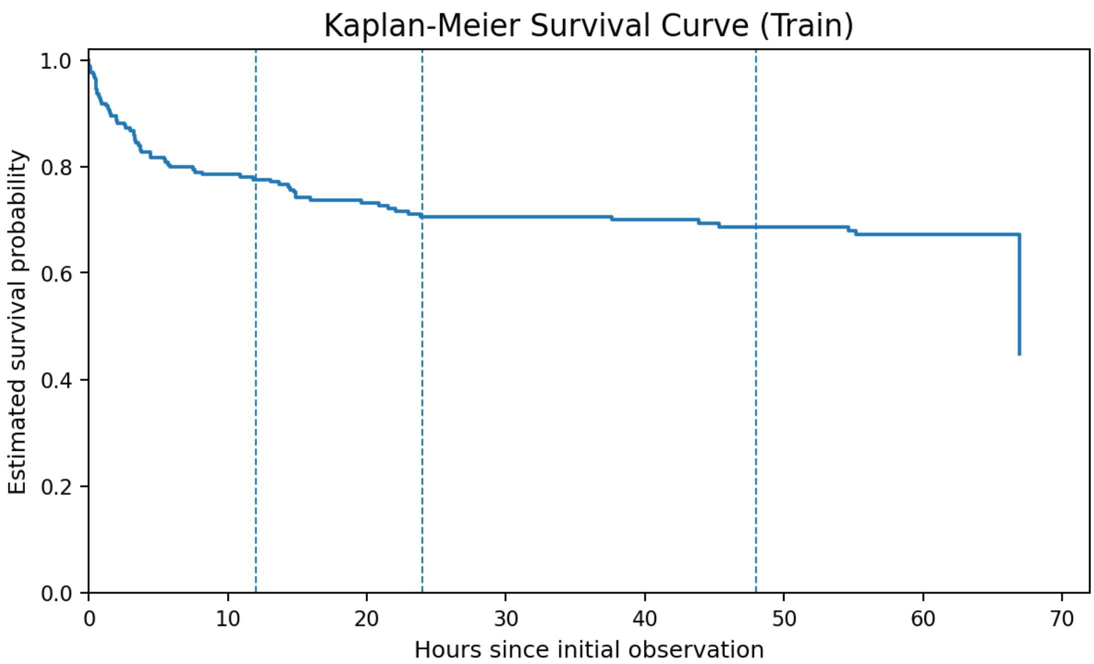

Figure 3 Kaplan-Meier survival curve: declines faster in the early stage, and then enters the long tail stage

Figures 1 to 3 show that this question has typical survival analysis characteristics: on the one hand, the missing samples account for the majority; on the other hand, the samples of real threats are highly concentrated in the earlier time window. The Kaplan-Meier curve declined rapidly at the beginning, indicating that once the fire developed towards a high-risk direction in the early stage, the threat would appear in a relatively short time; samples that did not enter the dangerous orbit in the early stage may remain alive until the end of the observation.

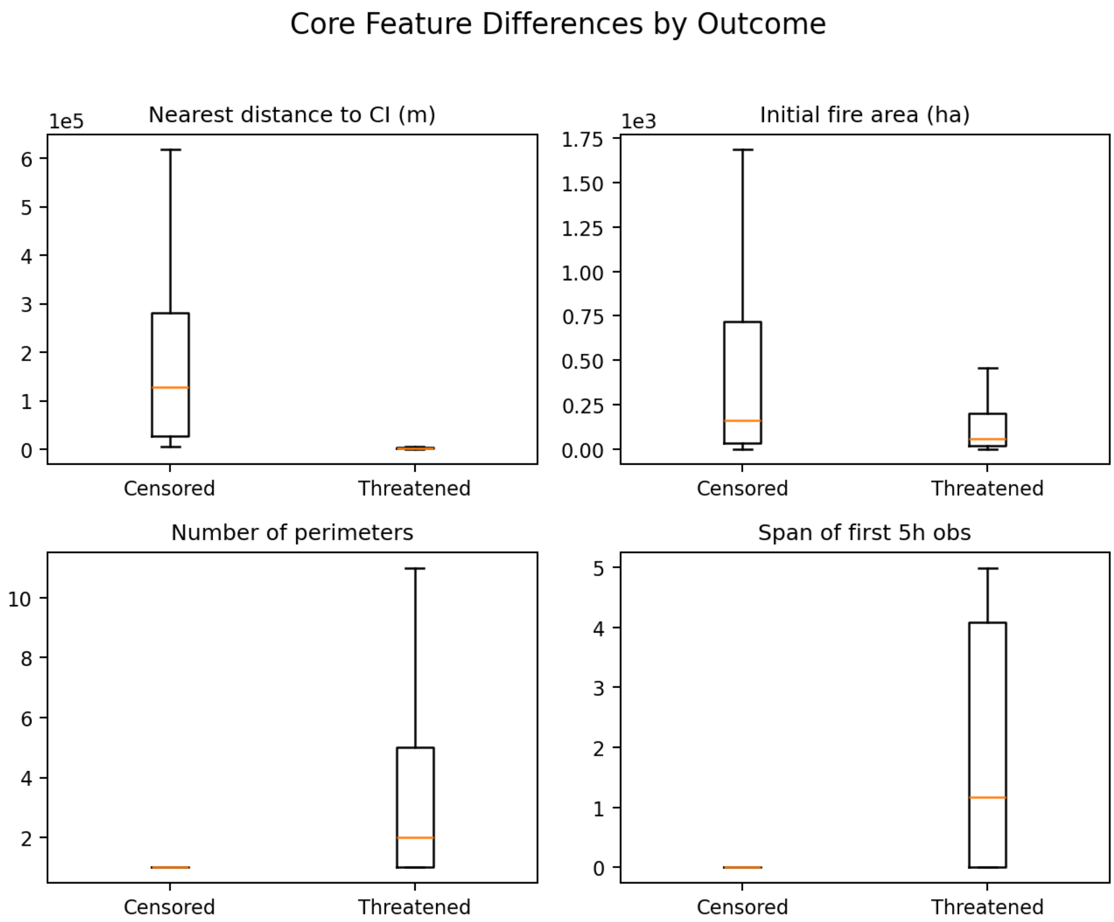

Figure 4 Differences in core characteristics in different result groups: distance, initial area, number of observation contours and observation span show obvious distinction

From the core characteristics, whether there is a threat is closely related to the spatial distance and early fire activity. dist_min_ci_0_5h is larger overall in the sample without threats, indicating that the fire field farther away from the target area is more difficult to form an actual threat within 72 hours; the threat sample usually has more early contour observations (num_perimeters_0_5h) and a longer observation span of the first 5 hours (dt_first_last_0_5h), indicating that the development process is more active and has more recognisable evolution trajectories.

There is a strong correlation between the variables related to area growth, which is also reflected in the relevant matrix diagram of the original EDA. High correlation shows that if it directly relies on simple linear models, the parameter interpretation may be unstable; tree models and regularised models are usually more advantageous in this scenario.

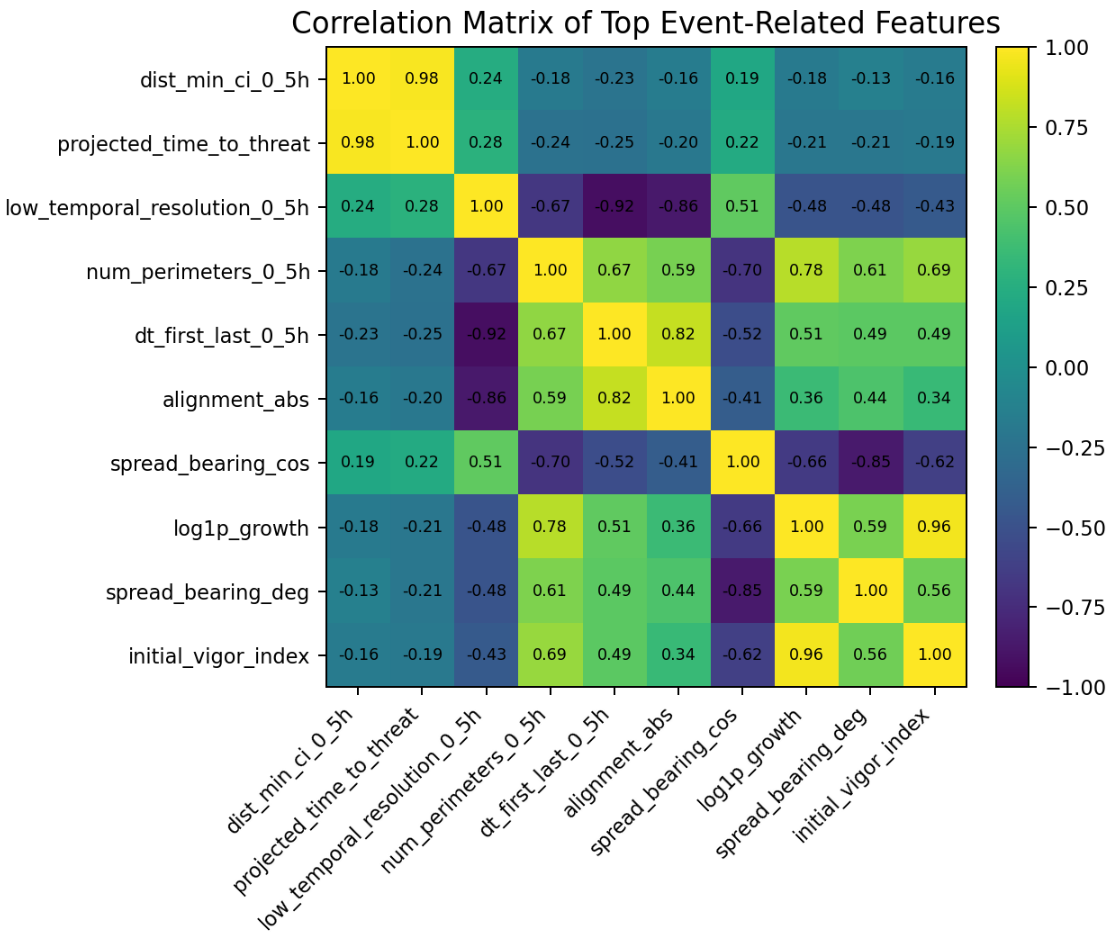

Figure 5 The correlation matrix of a group of characteristics with the strongest relationship with event: there is obvious collinearity between multiple groups of fire growth characteristics

In the feature engineering stage, 4 additional business-driven variables were constructed: projected_time_to_threat, kinetic_momentum, directed_speed_vector, initial_vigour_ Index.

Try to supplement the original characteristics from the four angles of "expected arrival time", "expanding kinetic energy", "moving speed along the target direction" and "initial fire vitality" respectively. According to the analysis results, the absolute value of the correlation coefficient between projected_time_to_threat and event is the largest, indicating that it does provide an additional signal that is highly consistent with "whether there is a threat"; initial_vigor_index also shows a certain Degree of distinction.

New Feature	Correlation coefficient with event	Correlation coefficient with time_to_hit_hours
projected_time_to_threat	-0.473	0.338
kinetic_momentum	-0.105	0.115
directed_speed_vector	0.008	-0.024
initial_vigor_index	0.268	-0.297

Table 3 Correlation between new features and target variables predictive modelling methods and results

4. The modelling work is divided into three levels: the first layer is to establish a reducible Cox baseline; the second layer is to compare multiple survival models on the enhanced feature set; the third layer is to perform Optuna parameterization after screening the candidate model to generate the final file in an integrated way.

4.1 Cox baseline model

The CoxProportional Hazards baseline model was established using the cleaned train_clean.csv, and Cox_Submission.csv was generated according to the survival probability of 12, 24, 48 and 72 hours. 00_metric_validation.ipynb scores the baseline prediction locally, and the output results are: C-index=0.8356, WeightedBrier=0.0814, Mixedscore=0.8937. There are two main meanings of the baseline model: first, the verification score and submission process are complete and operable; second, it provides a clear starting point for subsequent complex models.

4.2 Multi-model screening and internal benchmark comparison

In 04_model_training.ipynb, 9 types of survival models are compared on the enhanced feature set, including Cox series, tree model, upgrade model, SVM and XGBoost. Notebook uniformly outputs C-index, WeightedBrier and Mixedscore for the first round of candidate model screening. The score at this stage mainly comes from the internal Notebook output. The score is mainly used to quickly screen candidate models. It is not a strict final generalised evaluation result. It can be used as a reference for candidate sorting and does not directly represent the actual performance of the model on the external test set.

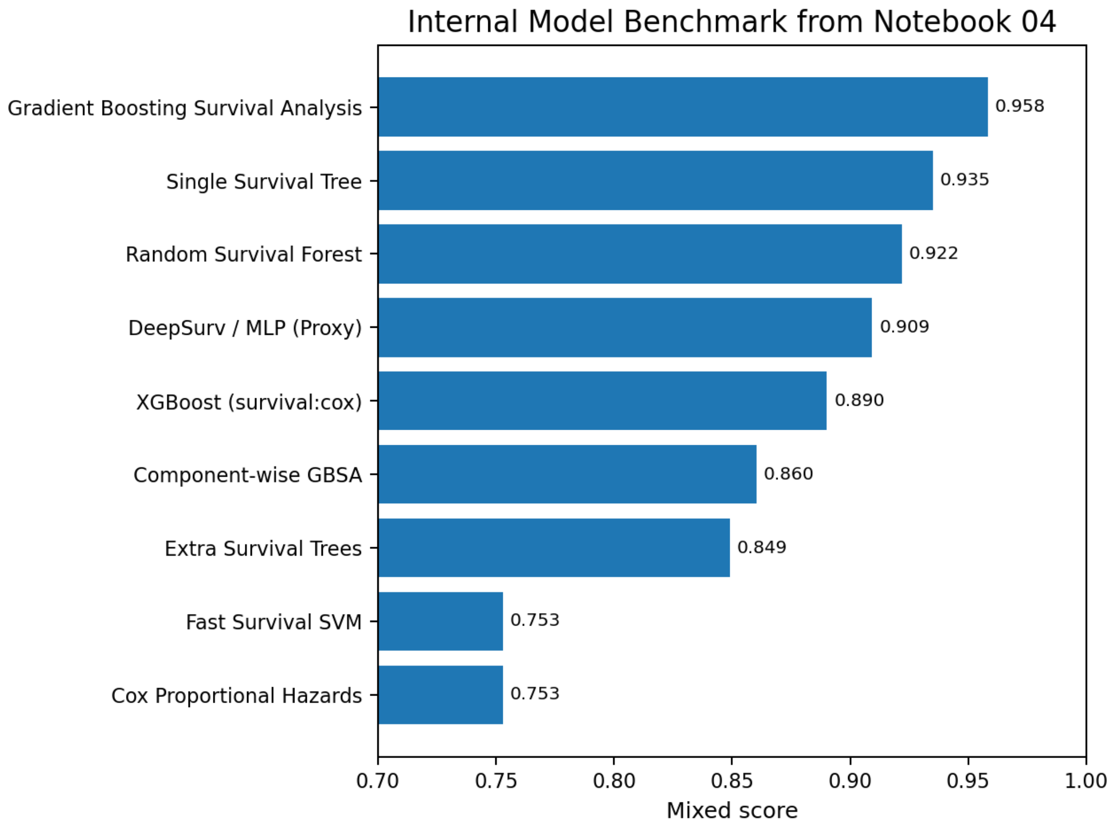

Figure 6 The internal model benchmark evaluation of Notebook 04 shows that GradientBoosting, SingleSurvivalTree and RSF are at the forefront.

Model	C-index	Weighted Brier score	Hybrid score
GradientBoostingSurvivalAnalysis	0.969	0.046	0.958
SingleSurvivalTree	0.950	0.072	0.935
RandomSurvivalForest	0.927	0.081	0.922
DeepSurv/MLP(Proxy)	0.904	0.088	0.909
XGBoost(survival:cox)	0.994	0.154	0.890
Component-wiseGBSA	0.889	0.152	0.860
ExtraSurvivalTrees	0.865	0.158	0.849
CoxProportionalHazards	0.904	0.312	0.753
FastSurvivalSVM	0.904	0.312	0.753

Table 4 Internal comparison results of multiple models

It can be seen from Table 4 that GradientBoostingSurvivalAnalysis obtained the highest mixed score of 0.958 in internal comparison; RandomSurvivalForest also maintained a high level; native XGBoost in C- The index is extremely strong, but the probability calibration part is relatively weak, so the comprehensive score is slightly lower than GBSA. The low mixed score of CoxPH and FastSurvivalSVM shows that relying solely on linear risk structures cannot fully capture the complex relationship in the competition. Tree models and upgrade models are more suitable for taking on the main role in terms of data.

4.3 Superparameter tuning and multi-model integration

05_hyperparameter_tuning_and_ensemble.ipynb uses Optuna to search for the best parameters for 9 models one by one. Under the framework of 30% cross-verification, with the combination of Brier and C-index in 48 hours los S as the search target. The reason why 48 hours is emphasised is that 48 hours has the highest weight in the official WeightedBrier, hoping to improve the sensitivity of the model to the critical time cross-section through settings.

After parameter adjustment, retrain each model on the full training set, and select 3 models that enter the final integration pool according to the C-index of the training set: NativeXGBoostSurvival, GradientBoostingSurvivalAnalys Is and CoxPHSurvivalAnalysis. Three types of files were generated: submission_ensemble_mean.csv, submission_ensemble_median.csv and submission_ensemble_rank.csv. The three correspond to arithmetic mean integration, median integration and rankaveraging integration respectively.

Selected Model	**Key Optimal Parameters**	**Full Training Set C-index**
NativeXGBoostSurvival	n_estimators=51;learning_rate=0.0161;max_depth=5	0.9762
GradientBoostingSurvivalAnalysis	n_estimators=77;learning_rate=0.1223;max_depth=3	0.9757
CoxPHSurvivalAnalysis	alpha=0.01146	0.9540

Table 5 3 models and optimal parameters of the final integrated pool

In terms of methodology, the three integration strategies have different divisions of labour: mean is suitable for smooth multi-model output, median is more robust for extreme prediction, and rankaveraging emphasises relative sorting, which is closer to the nature of C-index.

5. Kaggle ranking and final results

The latest publicleaderboard score of the team "IgnitionPredictors" is 0.96047, an increase from the previous version of 0.95271; the team ranking is 1013, and the external ranking score proves that the direction of multi-model integration is effective.

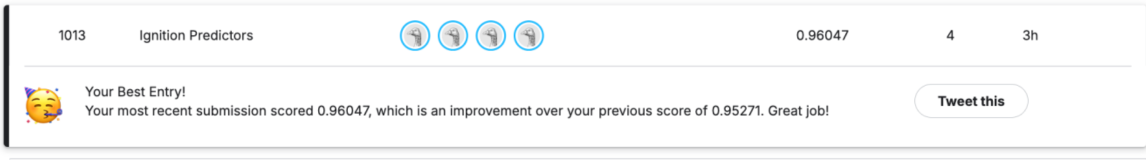

Figure 7 Kagglepublicleaderboard screenshot: the latest score is 0.96047, which is higher than the previous version 0.95271

The local Notebook score, internal screening score and Kagglepublicleaderboard score are not exactly the same in meaning. Local and internal scores are mainly used for a quick comparison in the research and development stage. Publicleaderboard is an external evaluation without test data, so the latter can better reflect the final value of the competition.

6. Method advantages, limitations and future work

First, it does not stay in a single model, gradually expands from the Cox baseline to 9 types of survival models, and finally enters the stage of parameter adjustment and integration, with a strong breadth of method exploration. Second, variables directly related to business logic are introduced in feature engineering, such as projected_time_to_threat, so that the model is closer to the actual problem of "when the evacuation area is threatened". Third, the improvement of the final publicscore shows that the integration and adjustment of participation are indeed effective in the performance of the competition.

Although it has achieved good results on the publicleaderboard, there are still certain limitations. There are only 221 samples in the training set, and the right deletion rate is close to 70%. The parameter estimates and generalisation conclusions of many models may be affected by small sample fluctuations. The internal comparison score of Notebook mainly comes from the internal evaluation of the training set or research and development stage. There is an optimistic bias, which cannot be directly equal to the final generalisation ability.

When adjusting the reference, the focus is on the 48-hour Brier. Although it conforms to the logic of the game weight, it may also weaken the overall calibration of 24 hours and 72 hours. Almost all of the current features come from the static or aggregate information of the first 5 hours, and the model does not make effective use of the finer-grained time series evolution and spatial propagation structure.

Follow-up improvement can be promoted from four directions. First, introduce stricter external cross-verification and repeated sampling evaluation to reduce the optimistic bias brought about by the internal screening process. Second, make a more systematic explanation of characteristics under the premise of ensuring no leakage, such as using permutationimportance or SHAP to help answer which early fire signals are most worthy of the attention of decision-makers. Third, combine finer-grained time series information, try DeepSurv, time-variingcovariates or sequence models to better simulate the evolution of fire over time. Fourth, further introduce spatial adjacency or geographical topology information, consider the spread relationship between different regions, and risk prediction does not only rely on single-point characteristics.

7. Group members and workload allocation matrix

Member	Student ID	Main Responsibilities
Zhang Xiao	25443771	Organizing competition background, providing literature support, writing the final report, and proofreading the final manuscript.
Zhong Liuyue	25428667	Responsible for data cleaning, exploratory data analysis (EDA), and feature engineering strictly based on the initial 5 hours of data.
Xu Jingyan	25435914	Responsible for building and tuning beginner-friendly models (e.g., Cox model and Random Survival Forest).
Zhu Yiwen	25435825	Responsible for monotonicity checks on

		predicted probabilities, calibration of results (optimizing Brier Score), code review, and final presentation video/report formatting.

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